

URBANGAZEVIS: A Visualization System for Analyzing Eye-Tracking Data on Urban Safety Perception

Graphical Abstract

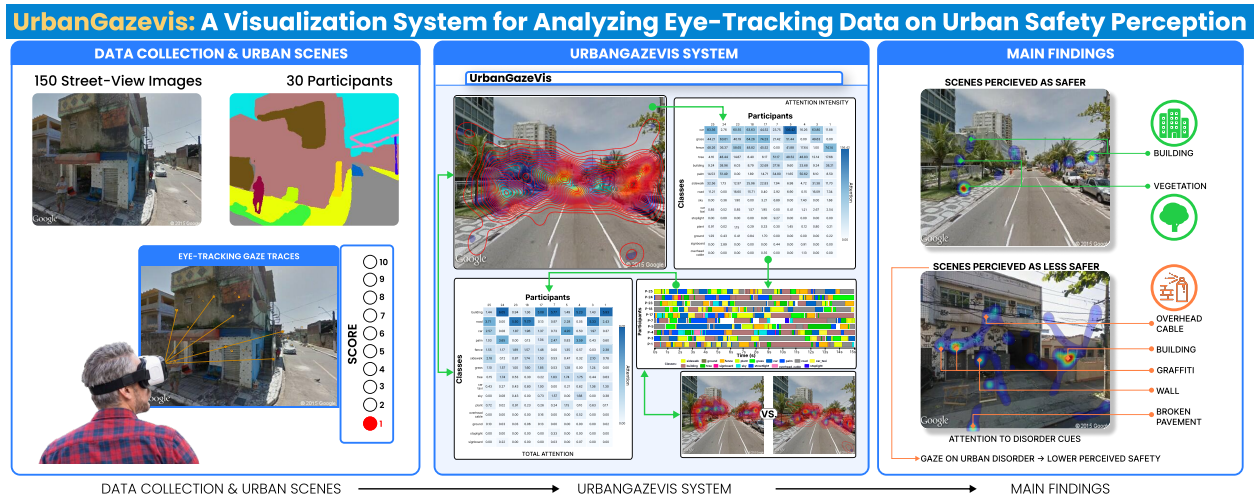


Figure 1: URBANGAZEVIS system overview and main findings organized into three components: Data Collection, URBANGAZEVIS System, and Main Findings. In the Data Collection stage, 30 participants viewed street-view images of urban environments while their eye movements were recorded using eye-tracking technology to capture gaze behavior during safety perception tasks. The URBANGAZEVIS System component visualizes gaze traces, attention intensity across semantic urban scene elements, and temporal patterns of visual attention, enabling the exploration and analysis of how participants interact visually with different environmental features. Finally, the Main Findings highlight that visual attention to urban disorder cues (e.g., graffiti, broken pavement, overhead cables) is associated with lower perceived safety, whereas attention to structured elements such as buildings and vegetation is linked to higher perceived safety.

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Highlights

- **VR-based eye-gaze collection methodology.** VR protocol for collecting and filtering gaze-based safety judgments.
- **UrbanGaze dataset.** Multimodal Rio street-view dataset with safety, gaze, and semantics.
- **UrbanGazeVis visualization system.** Interactive system linking gaze patterns to safety via semantics.
- **Main Findings:** Visual attention to urban disorder cues (e.g., graffiti, broken pavement, overhead cables) is associated with lower perceived safety.